

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 ( Supply. )**

**Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 7**

**Subject Code & Name: BTCVE705I - BRIDGE ENGINEERING**

---

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
- 

- Q1. Objective type questions. (Compulsory Question) 12
- 1 Bearings in a bridge are used to
    - a. Support the deck directly
    - b. Increase span length
    - c. Provide aesthetic appeal
    - d. Transfer loads from superstructure to substructure
  - 2 A bridge classified by location as “causeway” is built:
    - a. Across a deep navigable channel requiring high-level clearance
    - b. Along a steep embankment to connect two elevated hill slopes
    - c. Across shallow rivers with low embankments
    - d. Over a dry terrain where no water flow is expected
  - 3 Waterway calculation ensures:
    - a. Traffic volume management
    - b. Soil stability
    - c. Adequate width for flood flow passage
    - d. Proper deck thickness
  - 4 Freeboard in a river bridge is provided to:
    - a. Increase soil strength
    - b. Allow clearance above high flood level
    - c. Reduce scour depth
    - d. Increase deck thickness
  - 5 Longitudinal forces in road bridges are mainly due to:
    - a. Braking and tractive forces of vehicles
    - b. Wind loads
    - c. Dead load
    - d. Buoyancy

- 6 Impact factor accounts for:
  - a. Seismic forces
  - b. Dynamic effect of moving loads
  - c. Dead load
  - d. Buoyancy
- 7 Grillage analysis models the bridge deck as:
  - a. Hollow box section
  - b. Grid of beams
  - c. Prestressed slab
  - d. RCC wall
- 8 Box girder bridges are most suitable for:
  - a. Short spans only
  - b. Long spans with high torsional rigidity
  - c. Pedestrian bridges only
  - d. Temporary bridges
- 9 Parapets in highway bridges are mainly provided for:
  - a. Aesthetic appeal
  - b. Safety of vehicles and pedestrians
  - c. Reducing dead load
  - d. Increasing freeboard
- 10 Well foundations are commonly used for:
  - a. Small culverts
  - b. Temporary structures
  - c. Deep river crossings and heavy loads
  - d. Footpaths
- 11 Wing walls are provided to:
  - a. Increase foundation depth
  - b. Retain earth at bridge approaches
  - c. Reduce dead load
  - d. Increase pier height
- 12 The part of an abutment that resists horizontal earth pressure is:
  - a. Cap
  - b. Backwall
  - c. Toe slab
  - d. Curtain wall

Q2. Solve the following.

- A) Explain with neat sketch the components of a bridge. 6
- B) Explain the classification of road bridges based on span length. 6

Q3. Solve the following.

- A) Explain the factors affecting the selection of a road bridge site. 6
- B) State the factors governing the depth of bridge foundation. 6

Q4. Solve Any Two of the following.

- A) Define dead load and live load as per IRC specifications. How are they estimated for road bridges? 6
- B) What are the typical traffic loading patterns used for road bridges? 6
- C) Discuss the effect of centrifugal force and wind load on bridge structures. How are these loads considered in design? 6

- Q5. Solve Any Two of the following.
- A) What are expansion joints in bridges? Describe their necessity and various types used in modern bridge construction. 6
  - B) Define bridge parapet. Explain the classification and functional requirements of parapets for highway bridges. 6
  - C) Write short notes on:(a) Orthotropic plate theory (b) Grillage analogy 6
- Q6. Solve Any Two of the following.
- A) What are abutments? Describe their functions and the loads they resist. 6
  - B) State the importance of soil–structure interaction in bridge design. How does it affect foundation performance? 6
  - C) Explain the purpose of wing walls. How are they classified and designed? 6

**\*\*\* End \*\*\***